

Village Climate Report - Armoor, NAGARKURNOOL

Farm: RAMAVATH Family Farm · Armoor, NAGARKURNOOL, Telangana

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1. Farm Profile

Farm Name	RAMAVATH Family Farm
Owner	RAMAVATH VARUN
Village	Armoor
District	NAGARKURNOOL
State	Telangana
Area	12.5 Acres
Soil Type	Black Cotton
Irrigation	Borewell
Elevation	395.0 m

2. District Climate Risk Overview

OVERALL RISK 2.04 Low Risk	TEMP CHANGE +0.25°C by 2030 (SSP2-4.5)	RAINFALL CHANGE 1.29% projected shift	DROUGHT HAZARD 2.62 Low
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Flood Hazard	3.2 (Low)
Ag Stress	0.0 (Low)
Water Availability Index	1.02
Heat Stress Days	133.1

3. Climate Projections (SSP2-4.5, 2030 – 2040)

YEAR	OVERALL RISK	RISK CLASS	TEMP CHANGE	RAINFALL CHANGE	DROUGHT HAZARD	FLOOD HAZARD	AG STRESS
2030	2.04	Low Risk	+0.25°C	+1.29%	2.62	3.2	0.0
2035	2.01	Low Risk	+0.5°C	+2.54%	2.55	3.19	0.0
2040	8.1	Low Risk	+0.75°C	+3.79%	8.66	3.18	13.17

4. District Climate Insights

SSP245 – 2030:

- Temperature projected to increase by 0.2C by 2030.
- Agricultural stress remains Low.
- Rainfall change indicates a increase of 1.3%.

SSP245 – 2035:

- Temperature projected to increase by 0.5C by 2035.
- Agricultural stress remains Low.
- Rainfall change indicates a increase of 2.5%.

SSP245 – 2040:

- Temperature projected to increase by 0.7C by 2040.
- Agricultural stress is projected to rise sharply.
- Rainfall change indicates a increase of 3.8%.

SSP585 – 2030:

- Temperature projected to increase by 0.5C by 2030.
- Agricultural stress remains Moderate.
- Rainfall change indicates a increase of 1.3%.

SSP585 – 2035:

- Temperature projected to increase by 1.0C by 2035.
- Agricultural stress is projected to rise sharply.

- Rainfall change indicates a increase of 2.5%.

SSP585 – 2040:

- Temperature projected to increase by 1.4C by 2040.
- Agricultural stress is projected to rise sharply.
- Rainfall change indicates a increase of 3.8%.

5. Crop Suitability Projections — NAGARKURNOOL

CROP	CURRENT SUITABILITY	PROJECTED (2040, SSP2-4.5)	DECLINE CLASSES	RISK LEVEL	RECOMMENDATION
Cotton	Highly Suitable	Marginal	↓ 3 classes	Very High Risk	High risk under SSP585
Paddy	Suitable	Unsuitable	↓ 3 classes	Very High Risk	Shift area to short-duration paddy or millets
Maize	Moderate	Unsuitable	↓ 2 classes	High Risk	Limit to irrigated or low-risk blocks
Turmeric	Marginal	Unsuitable	↓ 1 class	Moderate Risk	Avoid expansion under high heat stress
Millets	Highly Suitable	Highly Suitable	— Stable	Low Risk	Prefer hardy short-duration varieties

6. Crop-Wise Detailed Analysis

Cotton

Current Suitability	Highly Suitable
Projected (2040)	Marginal
Risk Level	Very High Risk
Change Classes	3

⚙ **Recommendation:** High risk under SSP585

Paddy

Current Suitability	Suitable
Projected (2040)	Unsuitable
Risk Level	Very High Risk
Change Classes	3

⚙ **Recommendation:** Shift area to short-duration paddy or millets

Maize

Current Suitability	Moderate
Projected (2040)	Unsuitable
Risk Level	High Risk
Change Classes	2

⚙️ **Recommendation:** Limit to irrigated or low-risk blocks

Turmeric

Current Suitability	Marginal
Projected (2040)	Unsuitable
Risk Level	Moderate Risk
Change Classes	1

⚙️ **Recommendation:** Avoid expansion under high heat stress

Millets

Current Suitability	Highly Suitable
Projected (2040)	Highly Suitable
Risk Level	Low Risk
Change Classes	0

⚙️ **Recommendation:** Prefer hardy short-duration varieties

7. Strategic Interventions & Recommendations

Based on multi-decade climate modeling (SSP2-4.5 and SSP5-8.5 scenarios) for **NAGARKURNOOL** district, the following structural adjustments are recommended:

⚠ High-Risk Crops Detected

One or more crops in your district show significant suitability decline by 2040. Consider transitioning to more climate-resilient alternatives.

Water Management

- **Irrigation Infrastructure:** Given potential water availability changes, invest in drip or micro-sprinkler systems to optimize water use efficiency by up to 40%.
- **Rainwater Harvesting:** Construct farm ponds or check dams to capture monsoon surplus for use during dry spells.

Crop Diversification

- **Climate-Resilient Varieties:** Adopt drought-tolerant and heat-tolerant cultivars for crops showing suitability decline.
- **Crop Transition:** If any crop's projected suitability drops to "Unsuitable", phase into Millets, specialized Pulses, or other drought-resistant alternatives.
- **Intercropping:** Use mixed cropping systems to spread climate risk across multiple crop types.

Soil Health

- **Organic Carbon:** Increase soil organic carbon through mulching, cover cropping, and composting to improve natural water retention.
- **Conservation Tillage:** Reduce tillage intensity to preserve soil structure and moisture content.

Financial Risk Mitigation

- **Crop Insurance:** Enroll in PMFBY (Pradhan Mantri Fasal Bima Yojana) to cover climate-induced yield losses.
- **Diversified Income:** Explore agro-forestry, dairy, or apiculture as supplementary income streams to hedge against crop failures.

Disclaimer: This report is generated by FloodGuard Agri Intelligence using climate projection models (CMIP6 SSP scenarios), ICRISAT historical crop data, and geospatial analysis. Projections are indicative and based on best-available scientific data. Actual climate and agricultural outcomes may vary. This report does not constitute financial or agricultural advice. Consult local Krishi Vigyan Kendra (KVK) or agricultural extension officers for site-specific guidance.

Data Sources: CMIP6 Climate Models · ICRISAT District-Level Crop Database · India Meteorological Department · ISRIC SoilGrids · AWS Earth Search Sentinel-2 L2A.